

Lab 1: Film Ranking

Query

Query History

Scratch Pad

```

1  -- lab 1
2  WITH ranked_films AS (
3      SELECT
4          f.title,
5          c.name AS category,
6          f.length,
7          DENSE_RANK() OVER (
8              PARTITION BY c.name
9              ORDER BY f.length DESC
10         ) AS film_rank
11  FROM film f
12  JOIN film_category fc
13      ON f.film_id = fc.film_id
14  JOIN category c
15      ON fc.category_id = c.category_id
16  )
17
18  SELECT
19      title,
20      category,
21      length,
22      film_rank
23  FROM ranked_films
24  WHERE film_rank <= 3
25  ORDER BY category, film_rank, title;

```

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 64

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	title character varying (255) 🔒	category character varying (25) 🔒	length smallint 🔒	film_rank bigint 🔒
1	Darn Forrester	Action	185	1
2	Worst Banger	Action	185	1
3	Casualties Encino	Action	179	2
4	Quest Mussolini	Action	177	3
5	Gangs Pride	Animation	185	1
6	Pond Seattle	Animation	185	1
7	Sons Interview	Animation	184	2
8	Theory Mermaid	Animation	184	2
9	Lawless Vision	Animation	181	3
10	Conspiracy Spirit	Classics	184	1
11	Dracula Crystal	Classics	176	2
12	Lights Deer	Classics	174	3
13	Control Anthem	Comedy	185	1
14	Saturn Name	Comedy	182	2
15	Searchers Wait	Comedy	182	2
16	Caper Motions	Comedy	176	3
17	Sweden Shining	Comedy	176	3

Lab 2: Growth Analysis

```
27 -- lab 2
28 WITH monthly_revenue AS (
29     SELECT
30         DATE_TRUNC('month', payment_date) AS month,
31         SUM(amount) AS revenue
32     FROM payment
33     GROUP BY DATE_TRUNC('month', payment_date)
34 ),
35
36 revenue_with_previous AS (
37     SELECT
38         month,
39         revenue,
40         LAG(revenue) OVER (
41             ORDER BY month
42         ) AS previous_revenue
43     FROM monthly_revenue
44 )
45
46 SELECT
47     month,
48     revenue,
49     previous_revenue,
50     ROUND(
51         ((revenue - previous_revenue) / previous_revenue) *
52         100
53     ) AS growth_percent
54 FROM revenue_with_previous
55 ORDER BY month;
```

Data Output Messages Notifications

SQL

Showing rows: 1 to 4 of 1

	month timestamp without time zone	revenue numeric	previous_revenue numeric	growth_percent numeric
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.00
3	2007-04-01 00:00:00	28559.46	23886.56	19.56
4	2007-05-01 00:00:00	514.18	28559.46	-98.20

Lab 3: Customer Lifetime Value

```
57 -- lab 3
58 SELECT
59     customer_id,
60     payment_id,
61     payment_date,
62     amount,
63     SUM(amount) OVER (
64         PARTITION BY customer_id
65         ORDER BY payment_date
66         ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
67     ) AS cumulative_amount
68 FROM payment
69 ORDER BY customer_id, payment_date;
```

Data Output Messages Notifications

Showing rows: 1 to 1000 of 15

	customer_id smallint	payment_id [PK] integer	payment_date timestamp without time zone	amount numeric (5,2)	cumulative_amount numeric
1	1	18495	2007-02-14 23:22:38.996577	5.99	5.99
2	1	18496	2007-02-15 16:31:19.996577	0.99	6.98
3	1	18497	2007-02-15 19:37:12.996577	9.99	16.97
4	1	18498	2007-02-16 13:47:23.996577	4.99	21.96
5	1	18499	2007-02-18 07:10:14.996577	4.99	26.95
6	1	18500	2007-02-18 12:02:25.996577	0.99	27.94
7	1	18501	2007-02-21 04:53:11.996577	3.99	31.93
8	1	22680	2007-03-01 07:19:30.996577	4.99	36.92
9	1	22681	2007-03-02 14:05:18.996577	3.99	40.91
10	1	22682	2007-03-02 16:30:04.996577	0.99	41.90
11	1	22683	2007-03-17 11:06:20.996577	4.99	46.89
12	1	22684	2007-03-18 02:25:55.996577	0.99	47.88
13	1	22685	2007-03-19 08:23:42.996577	0.99	48.87
14	1	22686	2007-03-19 12:25:20.996577	2.99	51.86
15	1	22687	2007-03-21 22:02:23.996577	0.99	52.85

Lab 4: The Top N filter challenge

Step 1: Join 4 tables

```
72 -- lab 4
73 -- step 1
74 WITH customer_spending AS (
75     SELECT
76         c.customer_id,
77         c.first_name,
78         c.last_name,
79         co.country,
80         SUM(p.amount) AS total_spent
81     FROM customer c
82     JOIN address a
83     ON c.address_id = a.address_id
84     JOIN city ci
85     ON a.city_id = ci.city_id
86     JOIN country co
87     ON ci.country_id = co.country_id
88     JOIN payment p
89     ON c.customer_id = p.customer_id
90     GROUP BY
91         c.customer_id,
92         c.first_name,
93         c.last_name,
94         co.country
95 )
96 SELECT *
97 FROM customer_spending;
```

Data Output Messages Notifications

SQL Showing rows: 1 to 599

	customer_id integer	first_name character varying (45)	last_name character varying (45)	country character varying (50)	total_spent numeric
1	432	Edwin	Burk	South Africa	93.80
2	135	Juanita	Mason	India	100.72
3	543	Lance	Pemberton	Morocco	73.80
4	468	Tim	Cary	India	154.66
5	511	Chester	Benner	China	92.78
6	27	Shirley	Allen	Philippines	123.70
7	184	Vivian	Ruiz	Netherlands	80.80
8	102	Crystal	Ford	Israel	129.70
9	185	Roberta	Harper	Iran	79.80
10	530	Darryl	Ashcraft	Argentina	75.78
11	383	Martin	Bales	Angola	93.75
12	170	Beatrice	Arnold	India	109.77
13	75	Tammy	Sanders	Taiwan	149.61
14	32	Amy	Lopez	India	112.74
15	303	William	Satterfield	Yemen	70.81

Step 2: Apply RANK ()

```
-- step 2
ranked_customers AS (
    SELECT
        customer_id,
        first_name,
        last_name,
        country,
        total_spent,
        RANK() OVER (
            PARTITION BY country
            ORDER BY total_spent DESC
        ) AS spending_rank
    FROM customer_spending
)
```

Step 3:

```
111 -- step 3
112 SELECT
113     customer_id,
114     first_name,
115     last_name,
116     country,
117     total_spent,
118     spending_rank
119 FROM ranked_customers
120 WHERE spending_rank = 1
121 ORDER BY country;
122 ORDER BY country;
```

Data Output Messages Notifications

	customer_id integer	first_name character varying (45)	last_name character varying (45)	country character varying (50)	total_spent numeric	spending_rank bigint
1	218	Vera	Mccoy	Afghanistan	67.82	1
2	176	June	Carroll	Algeria	151.68	1
3	320	Anthony	Schwab	American Samoa	47.85	1
4	528	Claude	Herzog	Angola	93.80	1
5	381	Bobby	Boudreau	Anguilla	99.68	1
6	560	Jordan	Archuleta	Argentina	129.71	1
7	41	Stephanie	Mitchell	Armenia	118.75	1
8	266	Nora	Herrera	Austria	113.74	1
9	334	Raymond	Mcwhorter	Azerbaijan	108.75	1
10	590	Seth	Hannon	Bahrain	108.76	1
11	21	Michelle	Clark	Bangladesh	146.68	1
12	144	Clara	Shaw	Belarus	189.60	1
13	431	Joel	Francisco	Bolivia	92.78	1
14	178	Marion	Snyder	Brazil	194.61	1
15	91	Lois	Butler	Brunei	107.66	1
16	540	Tyrone	Asher	Bulgaria	112.76	1
17	516	Elmer	Noe	Cambodia	92.76	1